[INCH-POUND]
A-A-3038
October 10, 1996

COMMERCIAL ITEM DESCRIPTION

COMPRESSOR, PISTON RING, BAND TYPE

The General Services Administration has authorized the use of this Commercial Item Description for procurement of GGG-C-555 for all federal agencies.

1. SCOPE.

1.1. Scope. This Commercial Item Description covers piston ring compressors used to firmly compress the piston rings into the grooves of the piston to facilitate insertion of the piston and rings into the cylinder.

2. CLASSIFICATION.

2.1 Classification. Not applicable.

3. SALIENT CHARACTERISTICS.

3.1 Design. The piston ring compressors shall consist of a compression band, two tension bands, spool, ratchet, pawl and a removable key or wrench and shall be similar to Figure 1. The piston ring compressors shall have two size capacities; 2-1/8 to 5-inch, and 3-1/2 to 7-inch. The piston ring compressor shall be designed to accomplish the following:

a. To compress the heaviest piston rings serviced by means of twin band tension providing equalized pressure over the width of the compression band.
b. To provide the compression force by means of a spool, actuated by either a removable key or wrench.
c. To lock in position at any desired point within the operating range by means of a ratchet and pawl.
d. To provide take-up tension of approximately 0.010 inch per tooth on the ratchet.
e. To provide a friction brake for controlling the opening feature.
f. To provide a suitable means on the lower edge of the compression band from entering the cylinder when the piston is inserted.

3.1.1 Compression band. The compression band shall be of good quality spring steel, properly tempered so as to fully open the completely assembled compressor to the largest required diameter when not restrained. The width of the compression band shall not be less than 3-1/2 inches.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center, Kansas City, MO 64131.

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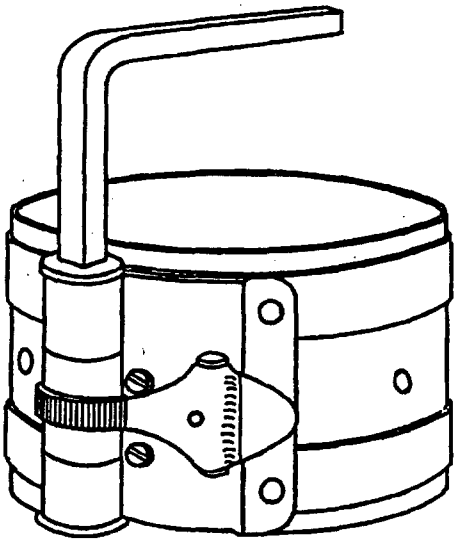


Figure 1. Compressor, piston ring, band type.

3.1.2 Tension bands. The two tension bands shall be of good quality spring steel, properly tempered, and of ample strength to compress the heaviest rings to be serviced. A means shall be provided for proper assembly of the tension bands to the spool.

3.1.3 Spool. The spool shall be of good quality steel, and be so designed to properly accept and wind the tension bands without slippage. A means shall be provided in the spool to properly accept a removable key or wrench which actuates the spool.

3.1.4 Ratchet. The ratchet shall be of good quality steel and properly heat-treated to resist wear or breakage under normal service conditions. The ratchet shall be designed for proper assembly to the spool and have the teeth so spaced as to accomplish a take-up of approximately 0.010 inch per tooth.

3.1.5 Pawl. The pawl shall be of good quality spring steel and properly heat-treated to resist wear or breakage under normal service conditions. The pawl shall be designed for proper assembly to the tension bands and shall lock the ratchet at the desired point.

3.1.6 Removable key or wrench. The removable key or wrench provided to actuate the spool shall be of good quality steel and of sufficient length and strength to actuate the spool in contracting the bands and the piston rings to the diameter desired.

3.2 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.3 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

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4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The product provided to the Government shall conform to the contractually specified requirements. For product characteristics and requirements which are not contractually specified, the provided product shall conform to the producer's own drawings, standards, specifications, and quality assurance practices for the highest quality product commercially offered by the producer. If the product is not commercially offered by the producer, the best commercial practices shall apply. The Government reserves the right to require proof of such conformances prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics by the Government. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

6. PACKAGING.

6.1 Packaging. Requirements of preservation, packing, packaging, and marking of packages shall be as specified in the contract or order.

7. NOTES.

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents:

Federal Specifications and Standards: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

Federal Acquisition Regulations (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20402-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 43201-4606.

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7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) When applicable, the appropriate NSN(s).
(c) If bid samples are required, the number of samples required, and the subjective inspection requirements.
(d) If first article samples are required and the number of samples required.
(e) If lot sampling inspection is required and the applicable lot sampling requirements. For example,

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Inspection. Each sample tool shall be examined and tested for all salient characteristics. The Inspection Level shall be S-2 with an Acceptable Quality Level of 4.0, expressed in terms of defects per hundred units.

(f) What preservation, packing, packaging, and marking of packages are required.

7.3 National Stock Numbers (NSNs). The following list of NSNs corresponds to the compressor specified in this CID. This list may not be indicative of all possible NSNs associated with this CID.

NSN	SIZE
5120-00-250-6055	2-1/8 inch to 5 inch capacity
5120-00-223-8848	3-1/2 inch to 7 inch capacity

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions or notices to this Commercial Item Description until further notice.

PREPARING ACTIVITY:

GSA - FSS

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